6.5.1 Using X-rays

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	basic structure of an X-ray tube; components – heater, cathode and target metal	
(b)	production of X-ray photons from an X-ray tube	

(c)	X-ray attenuation mechanisms; simple scatter, photoelectric effect, Compton effect and pair production.	
(d)	attenuation of X-rays; $I = I_0 e^{-\mu x}$, where μ is the attenuation (absorption) coefficient	*M0.5, M3.11*
(e)	X-ray imaging with contrast media; barium and iodine	
(f)	computerised axial tomography (CAT) scanning; components – rotating X-tube producing a thin fan-shaped X-ray beam, stationary ring of detectors, computer software and display	
(g)	advantages of a CAT scan over an X-ray image.	